Diego Emilio Barrera Hernández

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EDUCATION

ITESM, Queretaro Campus Graduated in June 2024
Bachelor of Science in Computational Technologies Engineering *GPA 94/100* *GPA 3.7/4.0*

ITESM, Queretaro Campus (semester) *GPA 100/100* *GPA 4.0/4.0*
Min. in AI and Data Science

WORK EXPERIENCE

Amazon Mexico City, Mex
Software Developer Engineer I Sep 2024 – Present

- Extended subscription campaigns for Mexico and Brazil sellers, coordinating cross-team efforts and delivering a production-safe deployment despite testing environment outage.
- Designed and executed a production deployment strategy with rollback planning and monitoring, preventing $7M+ in potential losses.
- Led engineering investigation for KYC integration in the US→Mexico shipping lane, identifying a carrier flow mismatch and implementing new system logic, preventing $2.8B in potential losses.
- Resolved 80%+ of on-call tickets, maintaining service reliability for internal and external systems.

P/S/L Group Queretaro, Mex.
Application Developer Intern Feb – Jun 2024

- Fullstack development of pharmaceutical systems using PHP, TypeScript, JavaScript, and React.
- Achieved outstanding results in a software development project using Scrum methodology and AWS services.
- Correct employment of the workflow in application of CodeCommit from AWS and database management with SQL and NoSQL.

General Electric Queretaro, Mex
Aviation Systems Intern Apr – Dec 2023

- Programmed and tested automatic codes for aircraft control systems (E747 and E737) using Jira and DOORS.
- Implemented testing methodologies including black box and white box testing, adhering to Scrum and Lean practices.
- Adapted to team workflow, analyzing software requirements in IBM DOORS.

PROJECTS

Aviation systems forecast Queretaro, Mex
Artificial intelligence and machine learning model for General Electric to predict when an airplane will enter stable cruise Aug – Dec 2023

- AI/ML model to predict stable cruise phase using 36 GB of aircraft data.
- Achieved 86% balanced accuracy with XGBoost; used Python, Pandas, NumPy, TensorFlow.
- Performed data analysis and ETL processes; followed CRISP-DM methodology.

Program manager in programming department Queretaro, Mex.
The objective was to establish a programming department capable of providing technological solutions to a wide range of clients. Feb – Jul 2023

- Managed project for GNP and Michelin, driving growth and collaboration of my team using Jira and Scrum methodologies.
- Achieved level 2.7 of the CMMI model.
- Succeeded in deploying AWS with PostgreSQL, using TypeScript, React, GitHub, and Jest, exceeding client expectations.

AWARDS

VEX National robotics Championship 2024

- Achieved ranking **1st place nationally**, where I led the robot programming stage.

VEX Robotics world championship 2024

- Achieved **ranking 33rd globally out of 100 teams**, where I led robot programming stage.

SKILLS

PROGRAMMING LANGUAGES

4 years: C++, Python
3 years: JavaScript, TypeScript
1 years: PHP, R, C#, Kotlin, Java, C

TECHNOLOGIES

HTML, CSS, node.js, Express, SQL, MongoDB (NoSQL), Jest, Git, Amazon Web Services (AWS), VirtualBox, Bash, Android SDK, Docker, EC2, React, Flask, Github, TensorFlow, SKlearn, Pandas, Numpy, XGBoost, Tableau, Cloud Watch

Software and teamwork methodologies

SCRUM, Lean, CRISP-DM, CMMI, RACI Matrix, Kanban

LANGUAGES

English (C1), German (B1) and Spanish (Native)